

## Research Question

How do changes in salinity affect physiological processes in *Carcinus maenas* and *Hemigrapsus oregonensis*?

Alternative: changes in salinity affect physiological processes and the responses differ between *C. maenas* and *H. oregonensis*

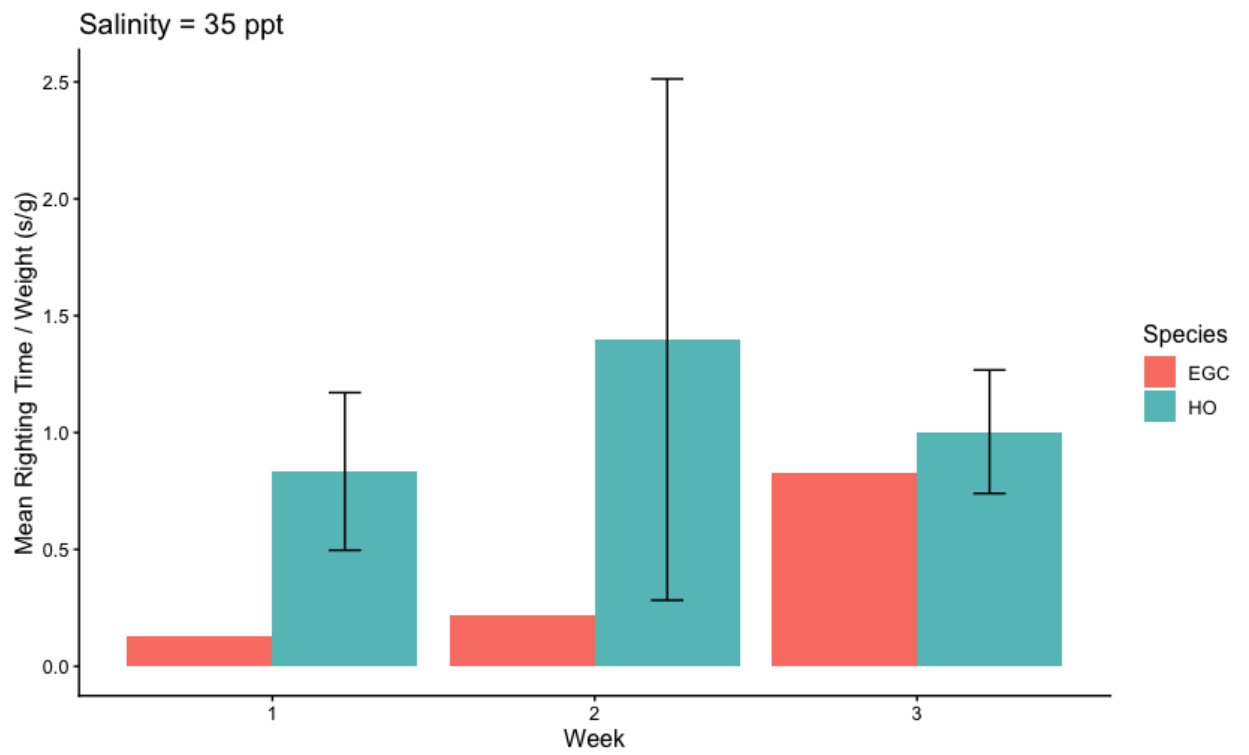
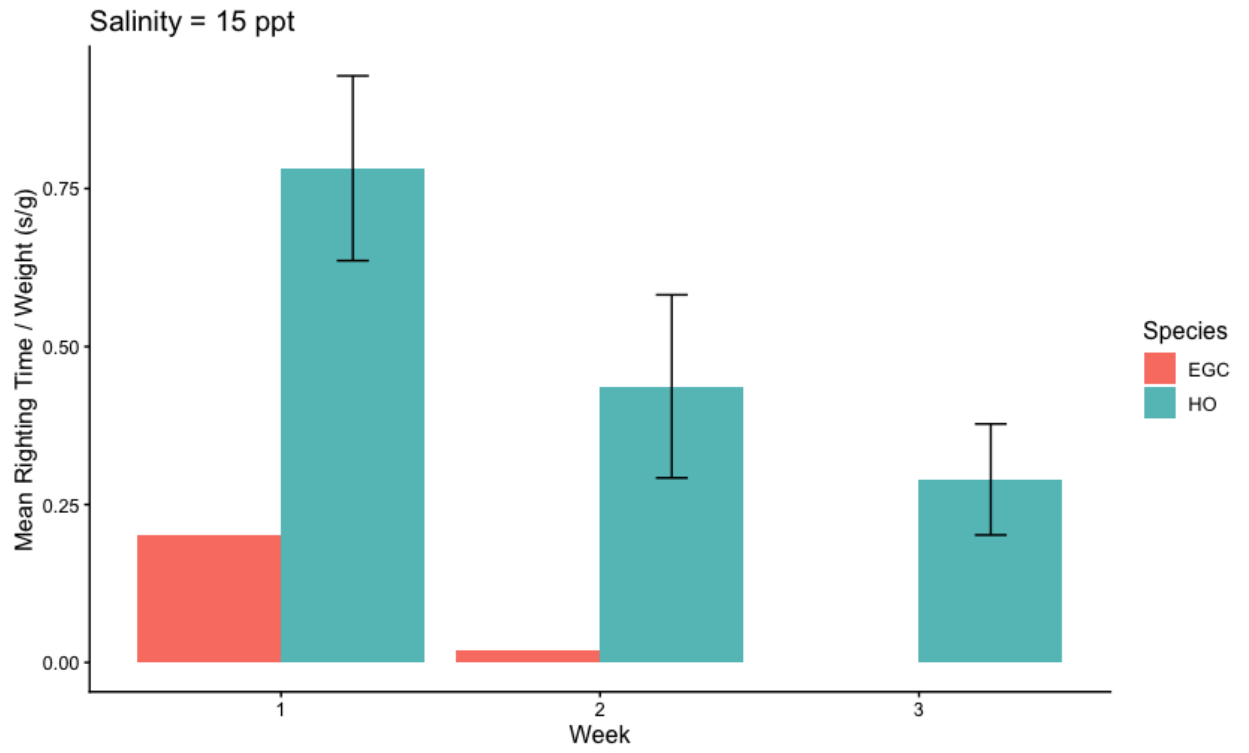
Null: changes in salinity have no effect on physiological processes, and there is no difference in physiological responses between *C. maenas* and *H. oregonensis*.

These crabs are well adapted to withstand wide shifts in salinity due to their environmental conditions in the intertidal zone. However, salinities may shift in the wild due to changing environmental conditions such as an increased influx in freshwater due to melting ice. This may introduce a small, long-term change in coastal salinity in Puget Sound with unknown effects of how these crabs will cope with such a change, and how their competitive nature may be altered. This study is designed to simulate salinity change extremes, allowing the observation of morphological changes over time, between salinity treatments, and between species.

## Hypotheses

- A lower salinity will result in an increased righting time as lower salinities may stress crabs out more than higher salinities since they are more likely to frequently experience high saline environments in the wild than low saline environments, which may increase stress and impact their ability to right themselves within a given time. Additionally, *H. oregonensis* will be able to right themselves faster than *C. maenas* due to their small anatomy and ability to manipulate their legs to a finer degree relative to *C. maenas*.
- A lower salinity will result in a reduced haemolymph osmolality. The crabs will be less able to cope with a reduced salinity for reasons listed in the previous hypothesis, and therefore will see the greatest deviation from baseline measurements. Additionally, when salinity is higher than typical environmental levels, *H. oregonensis* will have a higher osmolality than *C. maenas* and a lower osmolality relative to *C. maenas* when salinity is lower than typical environmental levels. Both of these hypotheses support the idea that *H. oregonensis* are less suited for coping with salinity changes.
- A lower salinity will result in increased respiration. The crabs will be more stressed in a reduced salinity environment for reasons stated in previous hypotheses, and therefore will consume a greater amount of oxygen/respire more as a result.

## Figures



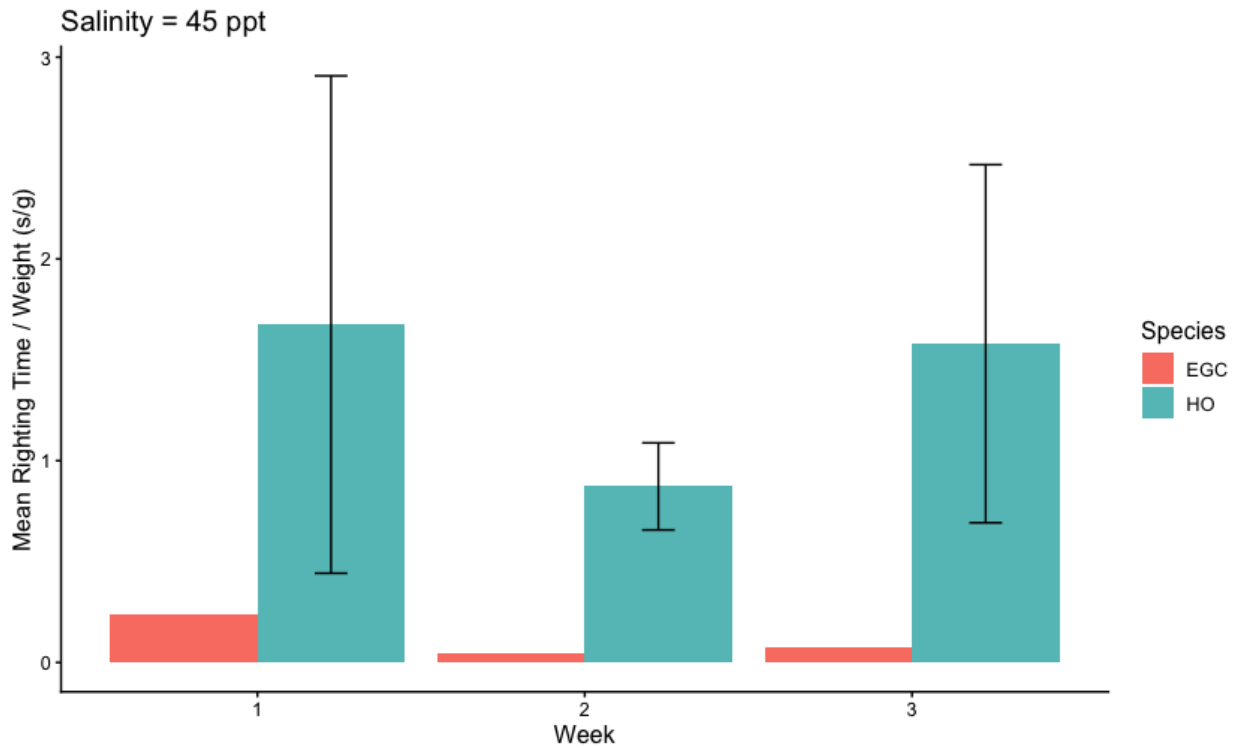


Figure 1 a, b, c - Mean righting time by mean weight of European green crab (EGC) and *Hemigrapsus oregonensis* (HO) at 15 ppt (Figure 1a), 35 ppt (Figure 1b), and Figure (1c).

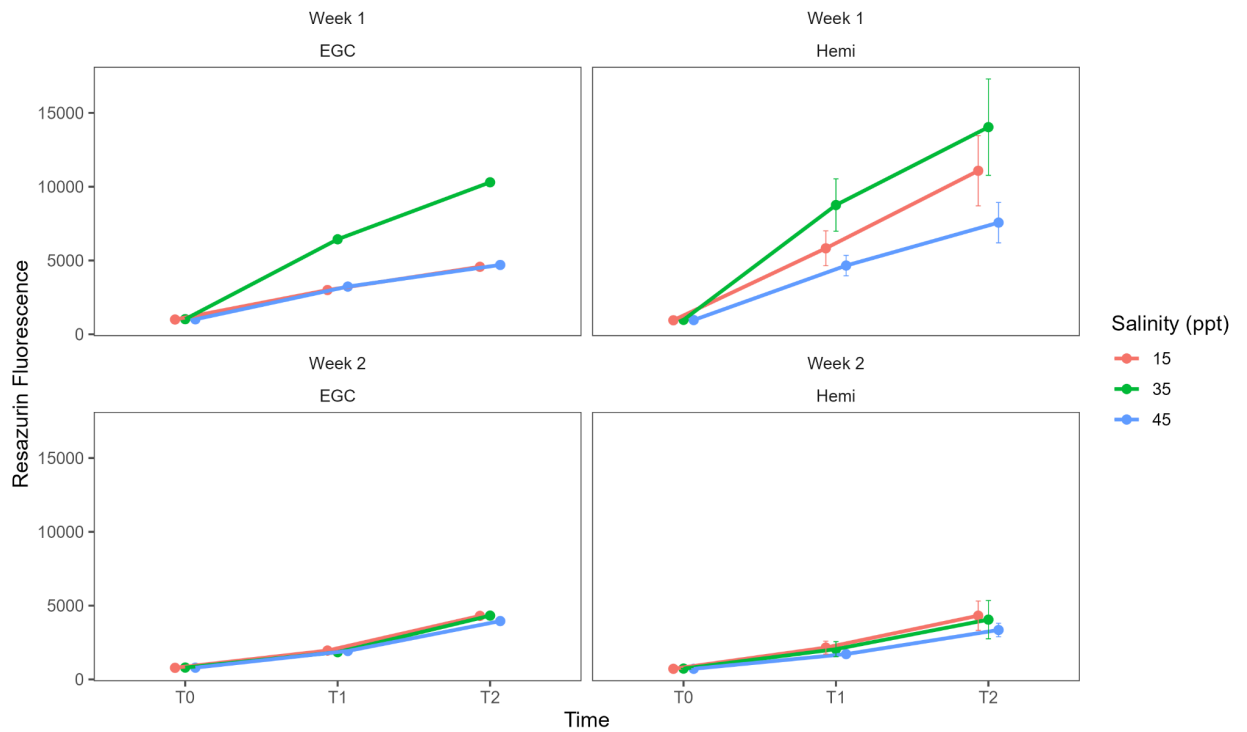


Figure 2. Hemigrapsus were submerged in fluorescence of resazurin, respiring over time. Fluorescence is a proxy for oxygen respiration. Each time step represents 30 minute intervals with T0 being the baseline.

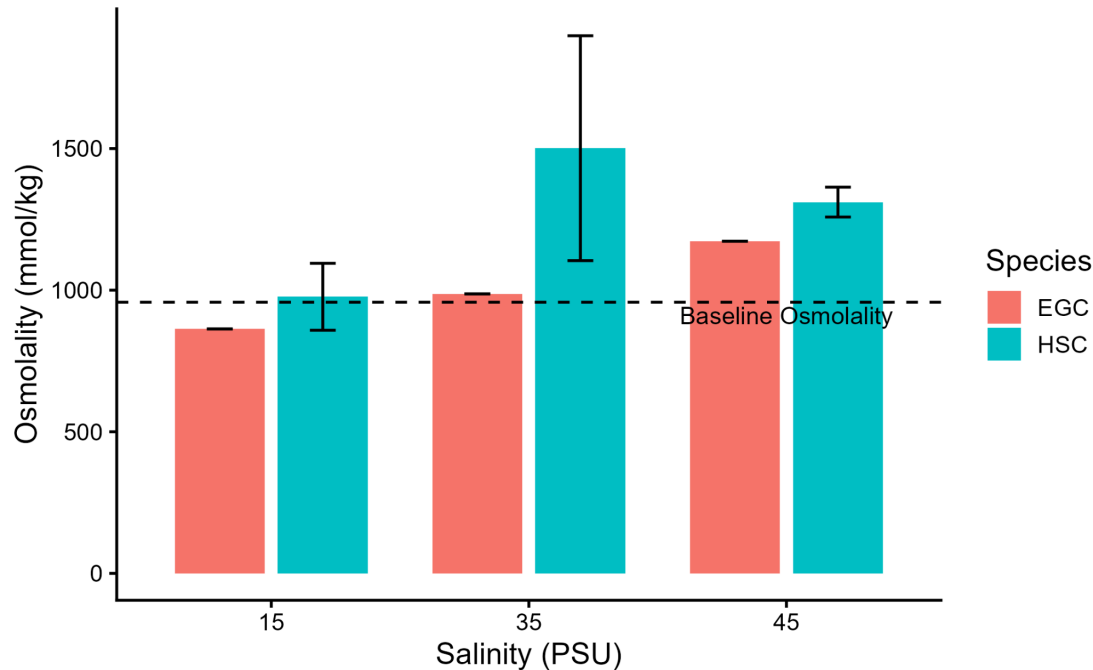


Figure 3. Osmolality (mmol/kg) of hemolymph after 3 weeks of sustained salinity treatments indicated on x-axis (15, 35, 45 PSU). Red bars indicate *C. maenas* (European Green Crab) and display one crab per treatment and therefore no standard error. Blue bars indicate *H. oregonensis* (Hairy Shore Crab) and display 3 crabs for 15 and 35 PSU, and 4 crabs for 45 PSU with standard error bars given those populations. Baseline osmolality measurement (dashed line) taken from 2 averaged, non-experimental *H. oregonensis* before the start of the experiment.

## Results

- Osmolality: A two-way ANOVA test was performed to assess the statistical difference between pairwise species-salinity tests. No statistical significance was observed between any species-salinity pairs. However, when each *H. oregonensis* salinity group was independently tested against the average baseline osmolality taken in week 1 with two-sided t-tests, crabs in the 45PSU treatment presented an osmolality significantly higher than the baseline ( $p\text{-value} = 0.006785$ ,  $df = 3$ ), while hairy shore crabs in the other salinity treatments were not significantly different than the baseline.

- Resazurin:

Table 1. Estimated Marginal Means (categorical version of slope).

| Treatment | Week | Species | emmean | SE   | DF | <a href="#">lower.CL</a> | <a href="#">upper.CL</a> |
|-----------|------|---------|--------|------|----|--------------------------|--------------------------|
| 15        | 1    | EGC     | 2857   | 946  | 4  | 231                      | 5483                     |
| 35        | 1    | EGC     | 5920   | 946  | 4  | 3294                     | 8546                     |
| 45        | 1    | EGC     | 2982   | 946  | 4  | 356                      | 5608                     |
| 15        | 2    | EGC     | 2349   | 75.2 | 4  | 2140                     | 2558                     |
| 35        | 2    | EGC     | 2324   | 75.2 | 4  | 2115                     | 2533                     |
| 45        | 2    | EGC     | 2226   | 75.2 | 4  | 2018                     | 2435                     |
| 15        | 1    | Hemi    | 5953   | 964  | 4  | 3277                     | 8629                     |
| 35        | 1    | Hemi    | 7924   | 964  | 4  | 5248                     | 10600                    |
| 45        | 1    | Hemi    | 4395   | 964  | 4  | 1719                     | 7071                     |
| 15        | 2    | Hemi    | 5953   | 964  | 4  | 3277                     | 8629                     |
| 35        | 2    | Hemi    | 7924   | 964  | 4  | 5248                     | 10600                    |
| 45        | 2    | Hemi    | 4395   | 964  | 4  | 1719                     | 7071                     |

## Interpretations

- Osmolality: The lack of significance within the ANOVA test indicates that salinity had no significant effect on osmolality regulation in either of these species, and neither of the species performed better or worse than the other. In comparison to the baseline osmolality measurement, 45PSU seemed to produce the greatest reaction within *H. oregonensis*, presenting a decrease in the ability to regulate osmolality, potentially expressing a proxy for excess stress.
- A significant amount of crabs died early in the study period due to an ammonia spike. This limited our sample size and reduced the amount of data for our analysis. A European green crab also molted, which likely affected respiration rates due to the increased physiological and metabolic demands.
- Righting time declined at 15ppt across both species of crabs, increased for EGC at 35ppt and no relationship for HO at 35, no relationship for EGC or HO at 45
  - HO always greater righting time than EGC, no relationship between the salinities
  - Crazy variance

